

Brac University
CSE427: Machine Learning
Fall 2025
Quiz 02

Duration: 20 minutes

Name:	ID:	Section: 06
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Question 1:

The initial Z value of the logistic regression classifier for an AND gate is $Z = -0.7X_1 + 1.2X_2 - 1$. Here X_1 and X_2 are the AND gate inputs. **Find** the cross entropy loss for each misclassified example assuming a probability threshold value equal to 0.4 . **Complete** one gradient descent cycle where the learning rate is 0.08 and update the Z whenever necessary.

CSE 427 Fall 25
Class Test 2
Time – 20 minutes

Section 2

ID:

Name:

The initial Z value for a logistic regression problem is $Z = 5X_1 + 2X_2 - 10$. For the following training set, find the cross entropy loss for each misclassified example assuming a probability threshold value equal to 0.6. Now iterate through the examples only once and update Z whenever necessary

X1	X2	Y
4	0	1
0	4	1
2	1	0
1	2	0